SIMATS ENGINEERING

Department of Computer Science and Engineering

CSA15 Cloud Computing and Big Data Analytics

CO2 AT1 - Capstone Infrastructure Sizing Workbook

Guided calibration workbook for VM sizing, virtualization decisions, snapshot planning and VM-container comparison

Assessment Metadata

Course Code	CSA15
Course Title	Cloud Computing and Big Data Analytics
Assessment	CO2 AT1 - Virtualization Scenario Problem-Solving
Submission Type	Individual digital workbook based on the same capstone title used in CO1
Faculty	Dr. C. Rajesh Babu
Employee ID	SSETSCS-415
Submission Mode	Completed workbook and evidence repository link through Google Classroom

How This Workbook Works

- Read each guidance block and immediately fill the related response table.
- Use the same capstone title selected for CO1.
- Make reasonable assumptions when exact values are unknown. Good assumptions must be written clearly.
- The goal is not to guess a perfect industry configuration; the goal is to reason technically and justify the first infrastructure plan.
- The final decisions from this workbook will be used again in CO2 AT2 and CO2 AT3.

Technical Tip

A good infrastructure estimate is transparent. If a number is assumed, write why it is assumed. If a number is calculated, show the formula. If a service is selected, justify why it fits the workload.

Student and Capstone Details

Student Name	G. Lakshmi Mounika
Register Number	192372110
Class / Slot	Slot B
Capstone Title	Cloud-Based CropWater AI: Irrigation Scheduling and Water Stress

Capstone Product Type	Web app / Android app / iOS app / Hybrid app
CO1 Architecture Reference	https://github.com/gundalakshmimounika/CSA1512-CLOUD-COMPUTING

Activity 1: Bring Forward CO1 Capstone Context

Start with the cloud product idea already used in CO1. The infrastructure plan must support that same application, not a new unrelated example.

Capstone Element	Student Entry
Primary users	Farmers, Agricultural Officers, Farm Managers
User roles	Admin, Farmer, Agriculture Expert
Main modules	Login, Dashboard, Farm Management, Soil Monitoring, Weather Monitoring, AI Prediction, Irrigation Scheduling, Reports
Cloud services identified in CO1	Firebase Authentication, Cloud Firestore, Firebase Storage, Firebase Hosting, Python AI API
API/backend need	REST API using Python Flask/FastAPI for AI prediction and irrigation schedulin
Database need	Cloud Firestore for storing user, farm, crop, weather and irrigation data
File upload/storage need	Firebase Storage for reports, farm images and datasets
Analytics/AI/Python module	Water Stress Prediction, Irrigation Recommendation, Crop Water Analytics

Activity 2: Workload Classification

Classify the capstone workload before selecting a VM. A form-based SaaS app and an AI analytics app should not receive the same infrastructure plan.

Workload Type	Typical Signals	Starting VM Guidance
Light web / form app	Login, CRUD, simple reports, low file upload	1-2 vCPU, 2-4 GB RAM
API backend	Authentication, REST APIs, moderate concurrent requests	2 vCPU, 4-8 GB RAM
Database-heavy app	Search, many records, frequent writes	2-4 vCPU, 8-16 GB RAM
File-heavy app	Images, PDFs, uploads, downloads	2 vCPU, 4-8 GB RAM + separate object storage
Analytics app	Dashboards, aggregation, batch reports	4 vCPU, 16 GB RAM
AI/Python module	Prediction, NLP, image/text processing	4-8 vCPU, 16-32 GB RAM; GPU optional
High traffic app	Large user base and peak demand	Multiple VMs or autoscaling group

Technical Tip

Start small for prototype, but do not hide growth. A valid answer can say: prototype uses one VM, pilot separates database, production uses load balancing and managed services.

Student Decision: My capstone workload type and reason
Workload Type: AI Analytics Web Application Reason: The project is a cloud-based web application that collects farm, soil and weather data, stores it in the cloud and uses Python AI models to predict water stress and recommend irrigation schedules. It requires cloud database services, analytics dashboards and AI processing.

Activity 3: User and Traffic Calibration

Use the following formulas to convert user assumptions into approximate traffic.

Metric	Formula	Example
Concurrent Active Users	Registered Users x Active User % x Peak Factor	500 x 10% x 2 = 100
Requests per Minute	Concurrent Active Users x Actions per User per Minute	100 x 2 = 200
Requests per Second	Requests per Minute / 60	200 / 60 = 3.33 RPS
Daily Requests	Daily Active Users x Avg Actions per Day	200 x 30 = 6000
Input	**Value**	**Reason / Assumption**
Registered users expected	500	Farmers, farm managers and agricultural officers using the system
Active user percentage	10%	Around 10% of users are active at peak time
Peak factor	2	Morning and evening irrigation create peak usage
Actions per user per minute	2	Viewing dashboard and checking irrigation recommendations
Calculated RPS	3.33 RPS	(500 × 10% × 2 = 100 concurrent users → 100 × 2 = 200 requests/min → 200/60 = **3.33 RPS**)
Daily active users	200	Approximately 40% of registered users access the sys
Daily requests	6000	200 daily users × 30 actions per day

Activity 4: CPU Calibration

CPU sizing depends on request rate and processing complexity. Use the score below as a guided estimate.

CPU Driver	Score
Base web/API application	1
Authentication and role management	+1
Moderate API traffic above 5 RPS	+1
Database-heavy search or reporting	+1 to +2

Background jobs or notifications	+1

Analytics or dashboards	+2
AI/Python inference inside server	+2 to +4

Total CPU Score	Recommended vCPU
1-2	1-2 vCPU
3-4	2 vCPU
5-6	4 vCPU
7-9	4-8 vCPU or separate analytics/AI service
10+	Scale out with multiple VMs or managed compute

CPU Driver Selected	Score	Reason
Base web/API application	1	Web application with dashboard
Authentication and roles	+1	Secure user login
API traffic	+1	REST API for prediction requests
Database/search/reporting	+1	Cloud Firestore and reports
Background jobs	+1	Scheduled irrigation notifications
Analytics/dashboard	+2	Dashboard and data visualization
AI/Python module	+3	Water stress prediction using Machine Learning
Total CPU Score	10	AI + Analytics + Cloud Application
Recommended vCPU	Multiple VMs or Managed Compute	Supports scalability and AI workloads

Activity 5: RAM Calibration

RAM must include operating system, runtime, application process, database/cache and a safety buffer.

RAM Formula

Estimated RAM = OS/runtime + app/API + database/cache + analytics/AI + 25% buffer. If database is managed separately, do not include full database memory in the app VM.

Component	Guidance	Student Estimate
OS + runtime	1-2 GB	2GB
Web/API service	1-2 GB	2GB
Small database	2-4 GB	4GB

Medium database/cache	4-8 GB	4GB

Analytics/reporting	4-8 GB extra	4GB
AI/Python module	4-16 GB extra	8GB
25% buffer	Subtotal x 0.25	6GB
Final RAM	Round up to standard size	32 GB RAM

Activity 6: Storage Calibration

Storage must cover application files, database records, uploads, logs and backup/growth buffer.

Storage Item	Formula / Guidance	Student Estimate
OS and application	20-40 GB	30 GB
Database	Avg record KB x records/day x days x 1.3 / 1024 MB	15 GB
File uploads	Avg file MB x files/day x days x 1.3	10 GB
Logs	Requests/day x log KB x days / 1024 MB	5 GB
Backup/growth buffer	Subtotal x 30%	18 GB(30%)
Final storage	Round up to standard disk size	80 GB SSD

Technical Tip

Do not store large uploads permanently inside the VM disk when object storage is available. VM disk is for OS and application. Object storage is better for images, PDFs, media and documents.

Activity 7: Select the VM Plan

Plan	When It Fits	Student Choice
Single VM prototype	Small demo, simple app, low traffic	NO
Two-tier design	App VM + managed/separate database	YES(Prototype)
Three-tier design	Frontend, backend/API and database separated	FUTURE
Containerized backend	Microservices, portable API, CI/CD	YES
Serverless function	Event-based task, notification, small automation	YES(Notifications & Alerts)
Managed database/storage	Reliability, backup and scaling required	YES(Firebase Firestore & Storage)

Final VM Plan: selected plan, vCPU, RAM, storage and reason
🎬 Plan: Two-tier Cloud Architecture 🎬 vCPU: 4 vCPU 🎬 RAM: 32 GB 🎬 Storage: 80 GB SSD 🎬 Reason: Supports web application, AI prediction, cloud database, analytics dashboard and future scalability.

Activity 8: Hypervisor and Virtualization Decision

Approach	Meaning	Best Use
Type 1 hypervisor	Runs directly on physical hardware	Production data center and cloud provider infrastructure
Type 2 hypervisor	Runs above an existing OS	Student lab, local testing, VirtualBox/VMware Workstation
Cloud-managed virtualization	Cloud provider hides hypervisor details	Most public cloud VM deployments
Container runtime	OS-level process isolation	Lightweight APIs, microservices, fast deployment
Context	**Selected Approach**	**Justification**
Local prototype/testing	Type 2 Hypervisor	Easy development using VirtualBox/VMware
Cloud pilot deployment	Cloud-managed Virtualization	Managed cloud infrastructure
Production growth scenario	Type 1 Hypervisor	Better performance and scalability
Module suitable for container	Docker Container	Python API and AI services are lightweight and portal

Activity 9: Snapshot and Clone Strategy

Snapshot protects a point in time. Clone creates a duplicate environment. Both are useful, but they are not the same.

Event	Snapshot or Clone?	Frequency / Timing	Retention / Reason
Before major deployment	Snapshot	Before every release	Quick rollback
Before database schema change	Snapshot	Before schema update	Protect database
Before OS/package update	Snapshot	Before update	Restore if update fails
Before review/demo	Snapshot	One day before demo	Stable demonstration
Create testing environment	Clone	When testing new features	Separate testing environment
Create team demo copy	clone	Before project presentation	Independent demo copy

Technical Tip

A snapshot is useful for rollback. A clone is useful for parallel testing. A clone should not be treated as the only backup.

Activity 10: VM vs Container Decision

Criteria	Virtual Machine	Container
Isolation	Strong OS-level isolation	Process-level isolation
Startup	Slower	Faster
Resource use	Higher	Lower
Best for	Full OS, legacy app, database VM	Microservices, APIs, portable services
Portability	Moderate	High
Student use	Clear infrastructure learning	Modern deployment learning

Capstone Module	VM / Container / Managed Service	Reason
Frontend	Managed Service (Firebase Hosting)	Fast and scalable hosting
Backend/API	Container	Easy deployment and portability
Database	Managed Service	Firebase Firestore provides backup and scaling
File storage	Managed Service	Firebase Storage is secure and scalable
AI/Python module	Container	Efficient deployment of AI model
Analytics/dashboard	Container	Supports analytics and reporting
Notification/background jobs	Serverless Function	Automatic alerts and scheduled tasks

Activity 11: Final Recommendation

Write the final infrastructure recommendation in 8-12 technical lines. Include VM size, hypervisor/virtualization approach, storage plan, snapshot/clone plan and VM-container decision.

Our Cloud-Based CropWater AI application is recommended to use a two-tier cloud architecture with 4 vCPUs, 32 GB RAM, and 80 GB SSD storage. Firebase Hosting will host the frontend, while Firebase Firestore and Firebase Storage will manage the database and uploaded files. The Python AI prediction service will run inside a Docker container for portability and easy deployment. Cloud-managed virtualization is selected for deployment because it simplifies infrastructure management and supports future scalability. Docker containers are preferred for the backend and AI module, while managed cloud services handle authentication, storage, and database operations. Snapshots should be taken before deployments, database changes, and system updates to enable fast recovery. Clones should be used to create separate testing and demonstration environments. This architecture provides secure, scalable, and cost-effective infrastructure suitable for a smart irrigation and water stress prediction application.

Carry-Forward to CO2 AT2 and CO2 AT3

Output from CO2 AT1	How It Will Be Used Later
VM sizing values	Used as configuration target in CO2 AT2 practical evidence
Hypervisor/virtualization choice	Used to justify deployment environment
Snapshot/clone plan	Used as backup and testing evidence
VM/container comparison	Used in CO2 AT3 infrastructure design case
Final recommendation	Used as the capstone infrastructure baseline

GitHub Submission Checklist

- Completed workbook file.
- README with capstone title, sizing summary and final recommendation.
- Optional architecture diagram or table exported as image/PDF.
- Repository link submitted in Google Classroom.